

Final ML Project Report: HR Analytics Job Change Prediction

Project Type: Classification

Goal: Predict whether a candidate is likely to look for a job change.

Approach:

The dataset was split into 70% training, 15% validation, and 15% test sets using stratification. Preprocessing was fit only on the training set to avoid data leakage. Numeric features were imputed and scaled. Nominal categorical fields were one-hot encoded. Ordered fields such as education level, experience, last_new_job, and company_size were converted into numeric meaning.

Required Models Trained:

Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, K-Nearest Neighbors, and Support Vector Classifier.

Model Selection:

The best validation model was BayesianRidge Probability Ensemble (Top 3) with validation ROC-AUC = 0.8066 and F1 = 0.5317.

The strongest final test ROC-AUC result was BayesianRidge Probability Ensemble (Top 3) with test ROC-AUC = 0.7922 and F1 = 0.5015.

Why ROC-AUC and F1 Matter:

Accuracy alone can be misleading because the target class is imbalanced.

ROC-AUC shows ranking quality across thresholds, while F1 balances precision and recall for the positive job-change class.

Ensemble Discussion:

The Average Ensemble combined the top 3 validation models by averaging their positive-class probabilities.

The BayesianRidge Probability Ensemble used the same top 3 model probabilities as inputs and learned regularized BayesianRidge blending coefficients from the validation set.

The ensemble comparison is useful because it shows whether combining models improves generalization or only adds complexity.

Final Recommendation:

Use the validation-selected model as the official final model, then discuss the test metrics as the unbiased final check.

If an ensemble performs better on both validation and test, it can be justified. If not, the simpler individual model should be preferred.